

UNIT 4: MOLECULAR GENETICS

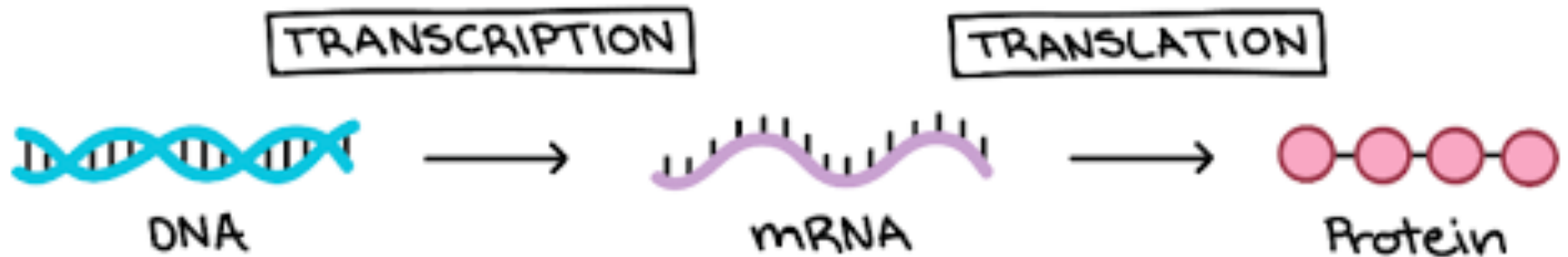
4.3 – DNA REPLICATION, TRANSCRIPTION, TRANSLATION

Biology 9

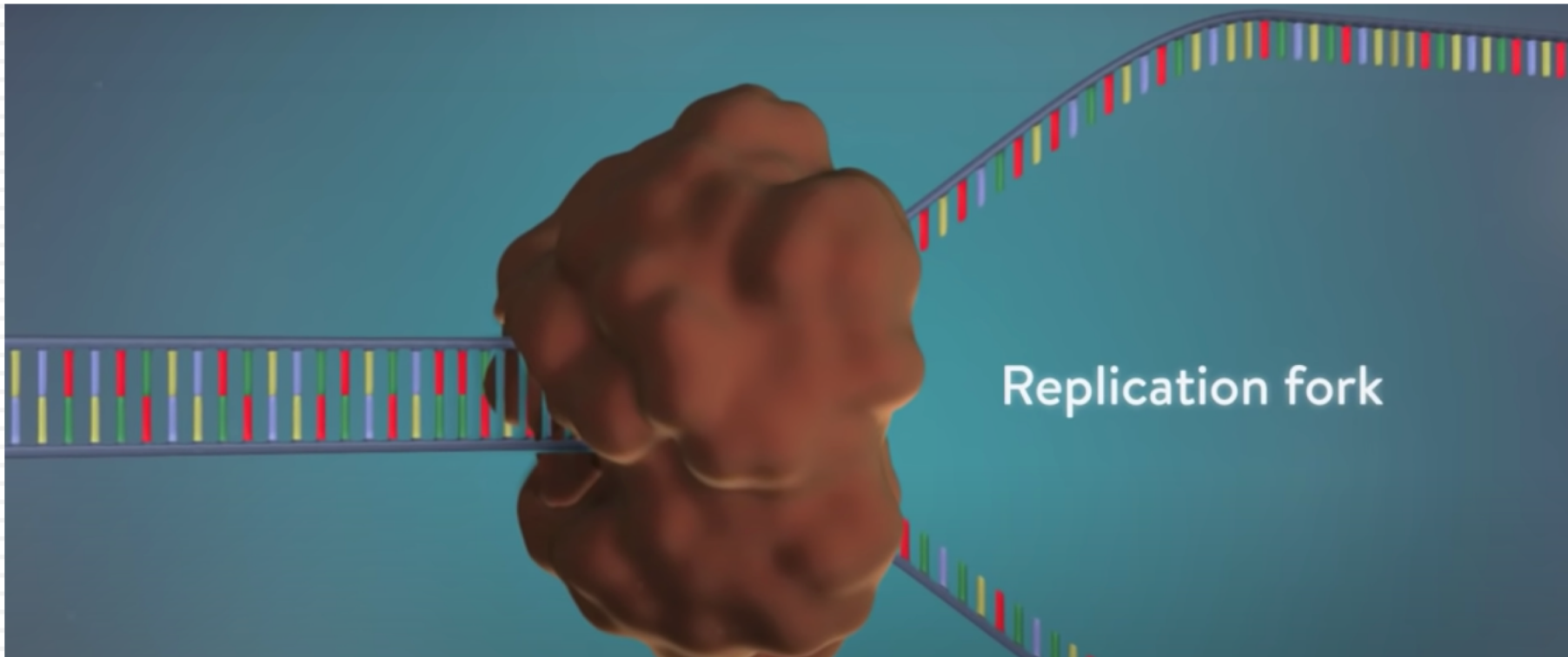
Mr. Queenan

Ch. 13.3 (p. 424-427) & Ch. 14.1-14.2 (p. 440-450)

Central Dogma of Biology



Replication of DNA



[Replication Video](#)

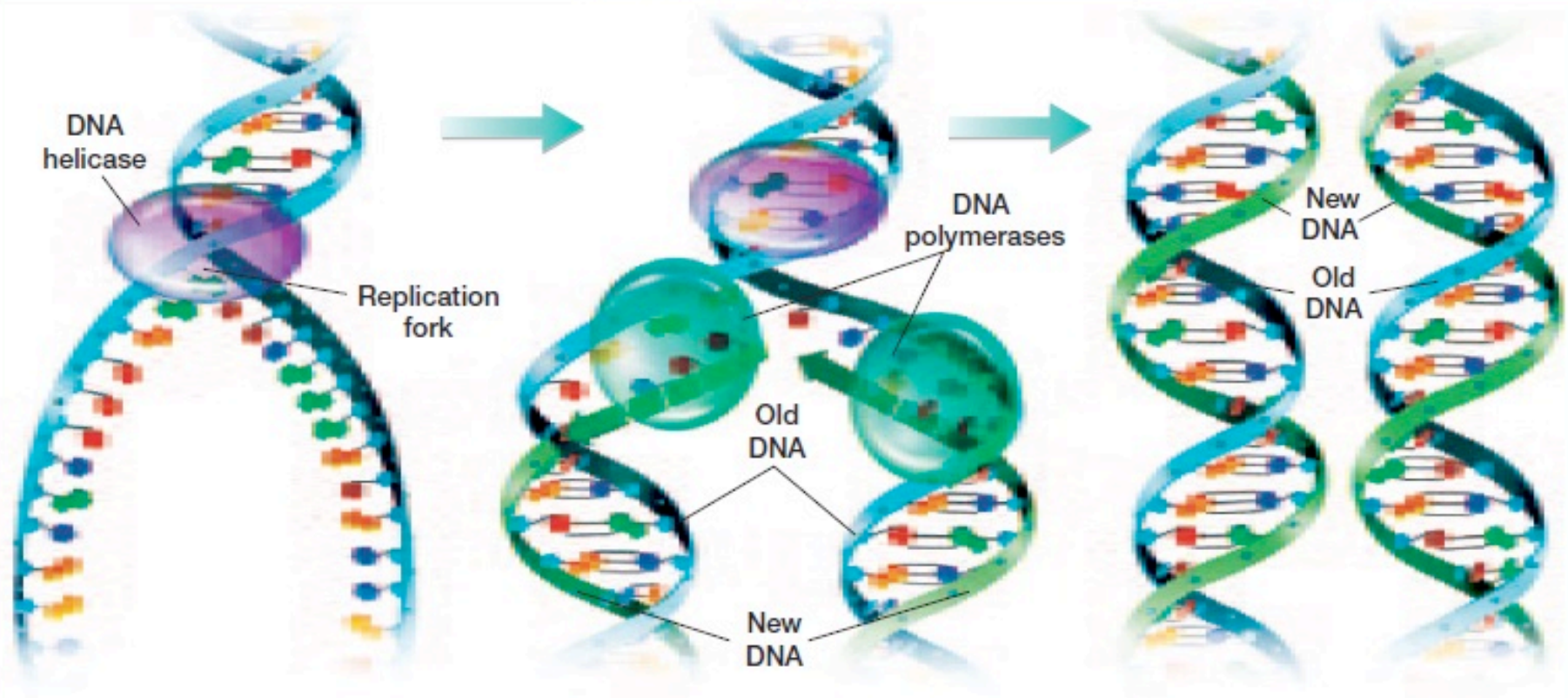
DNA Replication

- Process by which DNA is copied in a dividing cell
 - ▣ Mitosis, meiosis or binary fission
- The 2 nucleotide strands of the helix separate (melt)
- Complementary strands: each serves as a template for a new daughter strand
 1. **Helicases** separate/unwind the DNA strands
 1. Break the H-bonds between bases
 2. Y-shaped region results: replication fork
 2. **DNA polymerases** bind to DNA molecule, add complementary nucleotides, catalyze formation of covalent bond between nucleotides
 1. H-bonds spontaneously form between complementary bases between new and old strand
 3. DNA polymerases complete replication; fall off
- Result: 2 separate, identical DNA molecules ready for cell division

DNA Replication

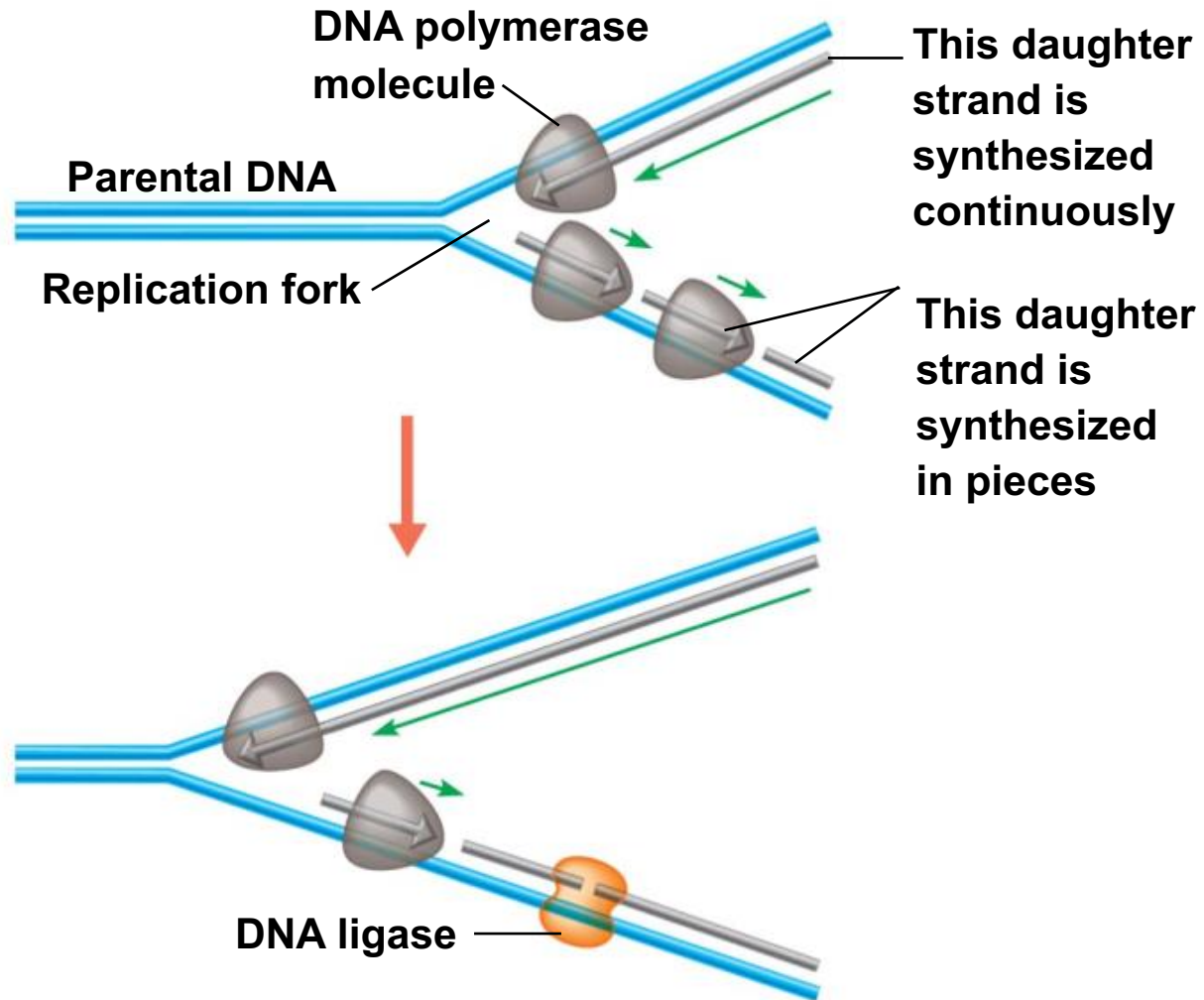
DNA replication results in two identical DNA strands.

- 1** The two original DNA strands separate.
- 2** DNA polymerases add complementary nucleotides to each strand.
- 3** Two DNA molecules form that are identical to the original DNA molecule.



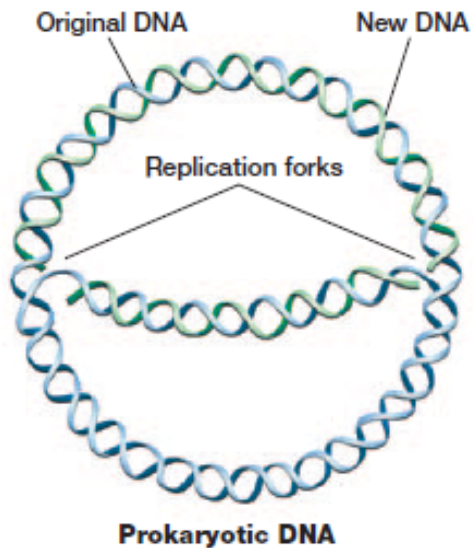
DNA Replication

- Each new DNA double helix: 1 strand from original molecule; 1 strand newly synthesized
 - ▣ Known as **semi-conservative replication**
- Synthesis occurs in different directions on each strand
- As replication fork moves along DNA template molecule, synthesis of one strand follows
 - ▣ Synthesis occurs from 5'-3'; each strand of DNA has a different directionality.
 - 5'-3', complement is 3'-5' when lined up next to each other
- Synthesis on other strand moves in opposite direction; away from replication fork; leaves gaps in new strand
 - ▣ Known as **Okazaki fragments**
 - ▣ Joined together by **DNA ligase**

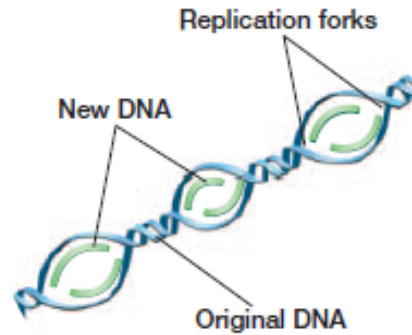


Prokaryotic vs Eukaryotic Replication

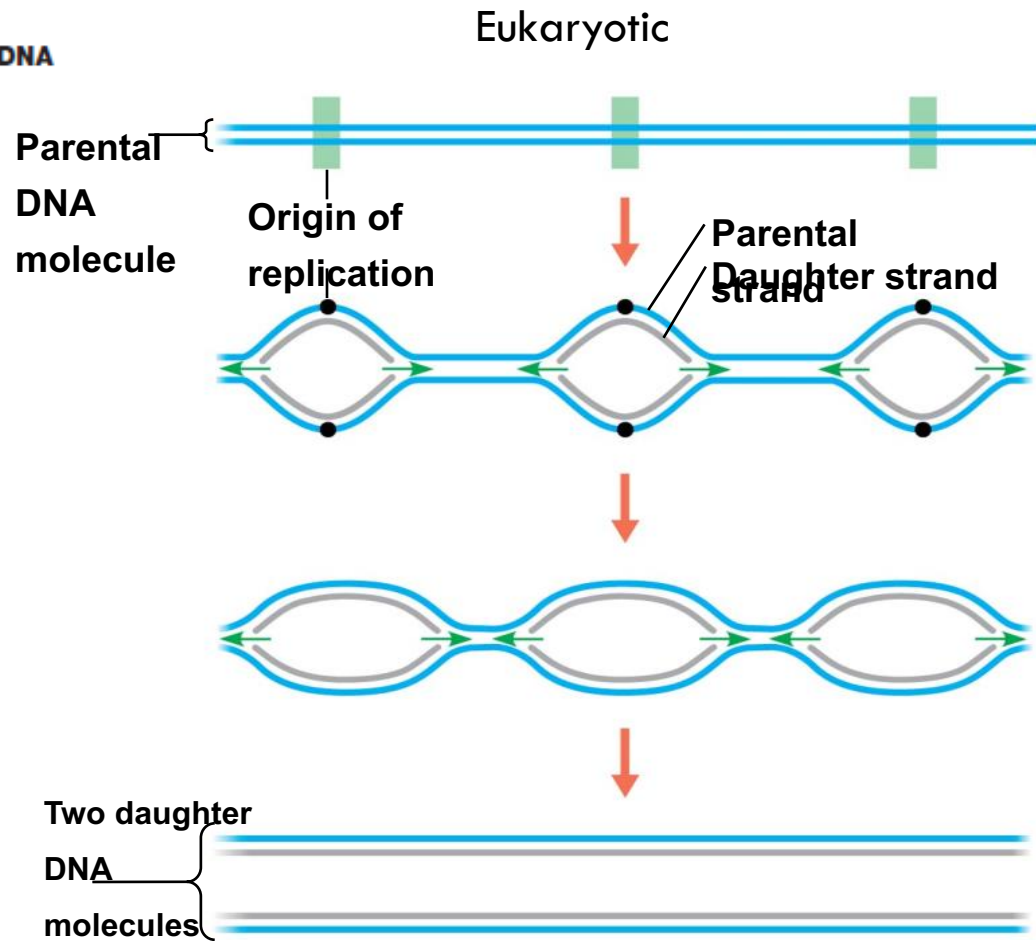
- Prokaryotes:
 - ▣ 1 circular chromosome
 - ▣ Replication begins at one place
 - ▣ 2 replication forks formed, move in opposite directions; continues until entire molecule is copied
- Eukaryotes:
 - ▣ Long, linear chromosomes (~3 billion base pairs in length)
 - ▣ Too long to have only one site of replication
 - ▣ Replication begins at many points along DNA
 - Known as **origin of replication**
 - Occur every few thousand base pairs



Prokaryotic



Eukaryotic DNA



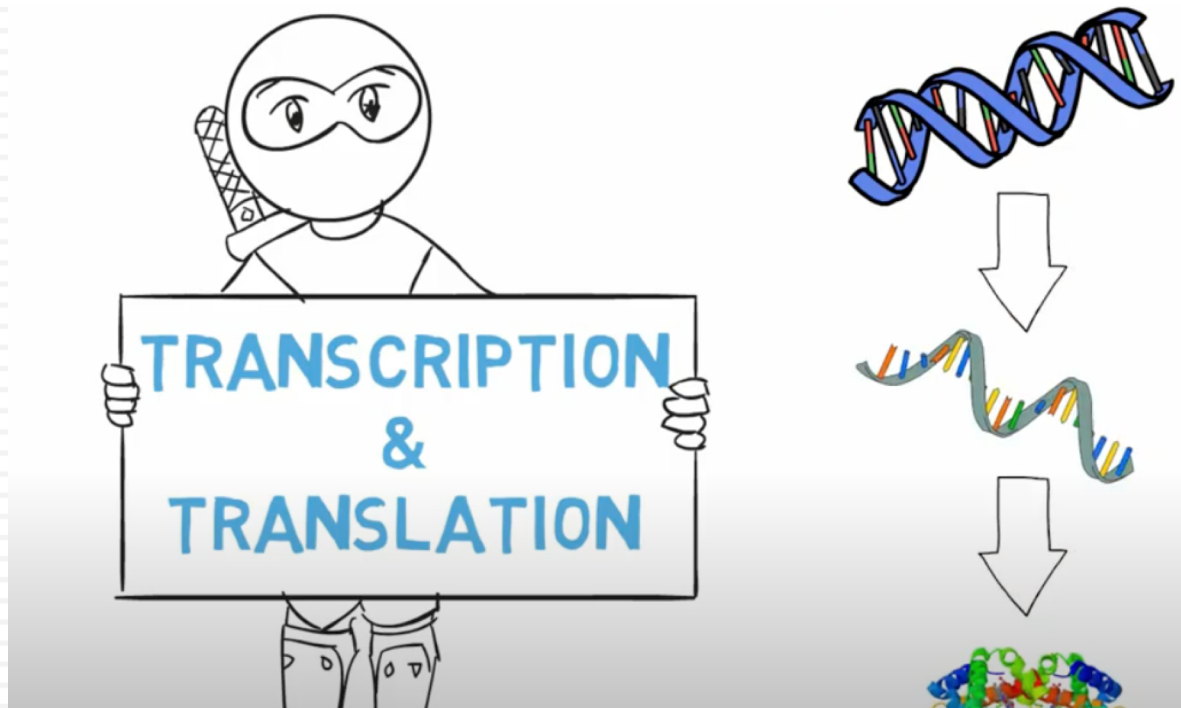
Errors in DNA Replication

- Extremely accurate process: average one error per billion nucleotides copied
 - ▣ How?
- DNA polymerase also has proofreading function
 - ▣ Reads newly synthesized DNA to make sure bases are paired correctly
 - ▣ When error does occur, incorrect nucleotide removed, correct one inserted
- Error in replication results in base sequence of new DNA different from parent DNA
 - ▣ Change in nucleotide sequence: mutation
 - ▣ Can change gene; disrupt cell function
- Not all errors can be repaired
 - ▣ Chemicals, UV radiation from sun cause DNA damage
 - ▣ Some mutations lead to cancer

Replication, Mutations & Cancer

- Most mutations in DNA occur during replication; sometimes other causes
- Some mutations lead to advantage for cell → adaption → evolution
- Some mutations cause no change in phenotype → silent mutation
- Some mutations deleterious:
 - ▣ Mis-functioning protein (disease)
 - ▣ Non-functional protein (disease)
 - ▣ Mutation in cell cycle regulatory genes (tumor suppressor/oncogene) lead to cancer

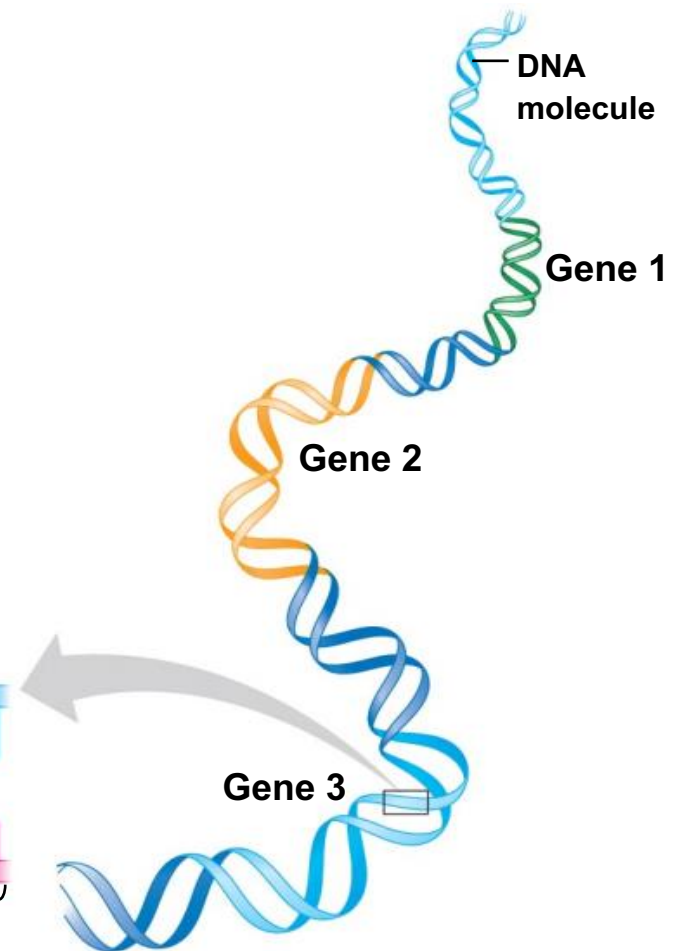
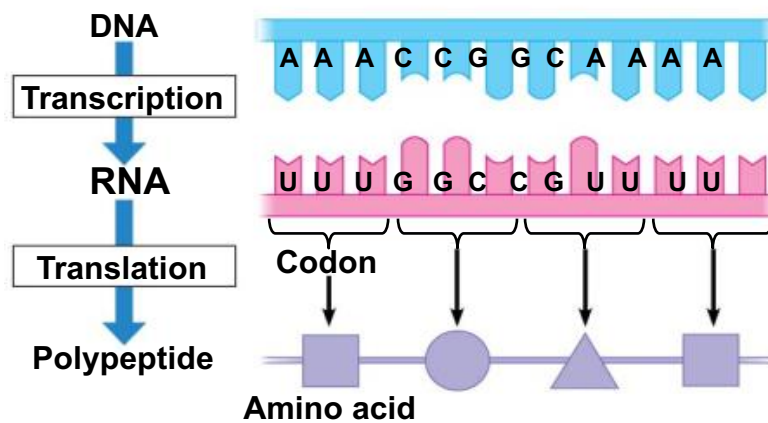
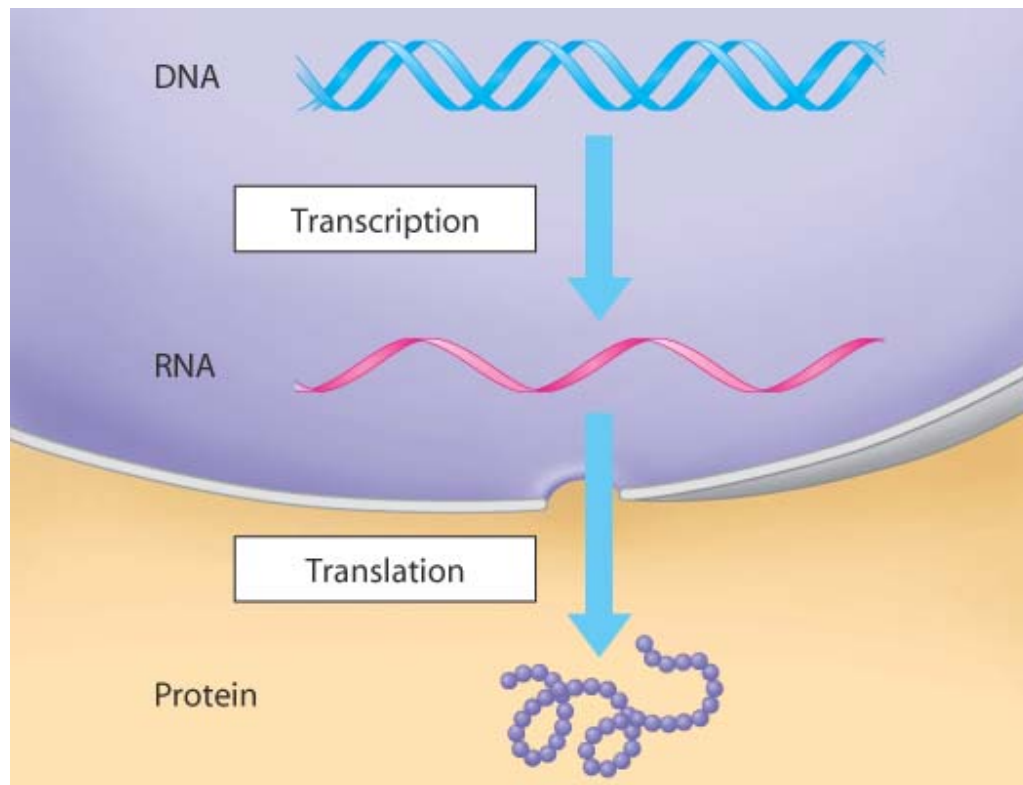
Protein Synthesis – Transcription & Translation



[Transcription/Translation Video](#)

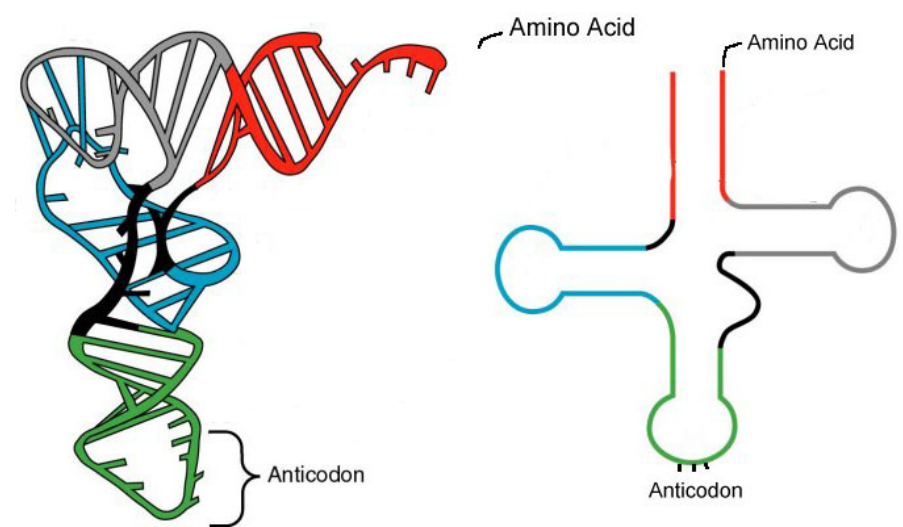
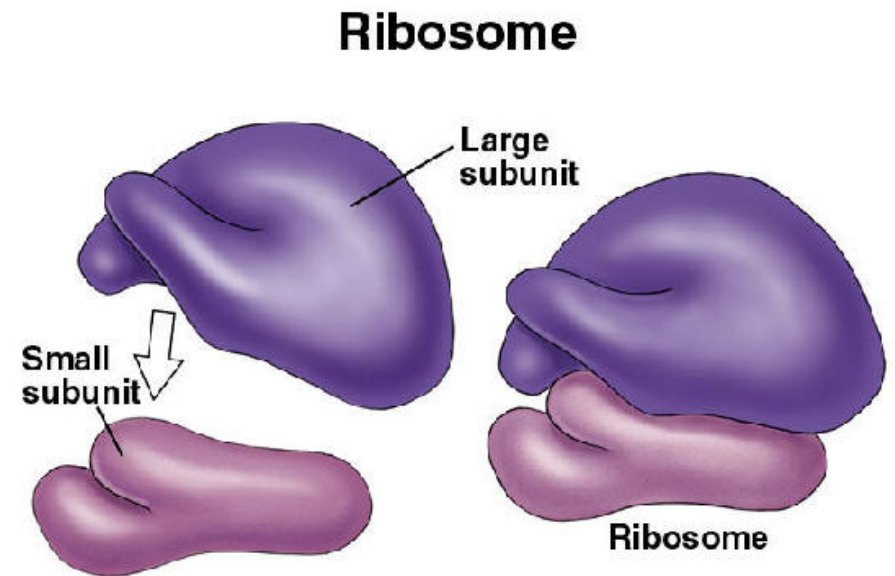
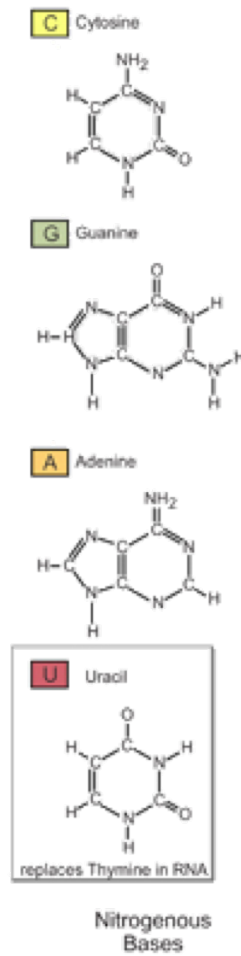
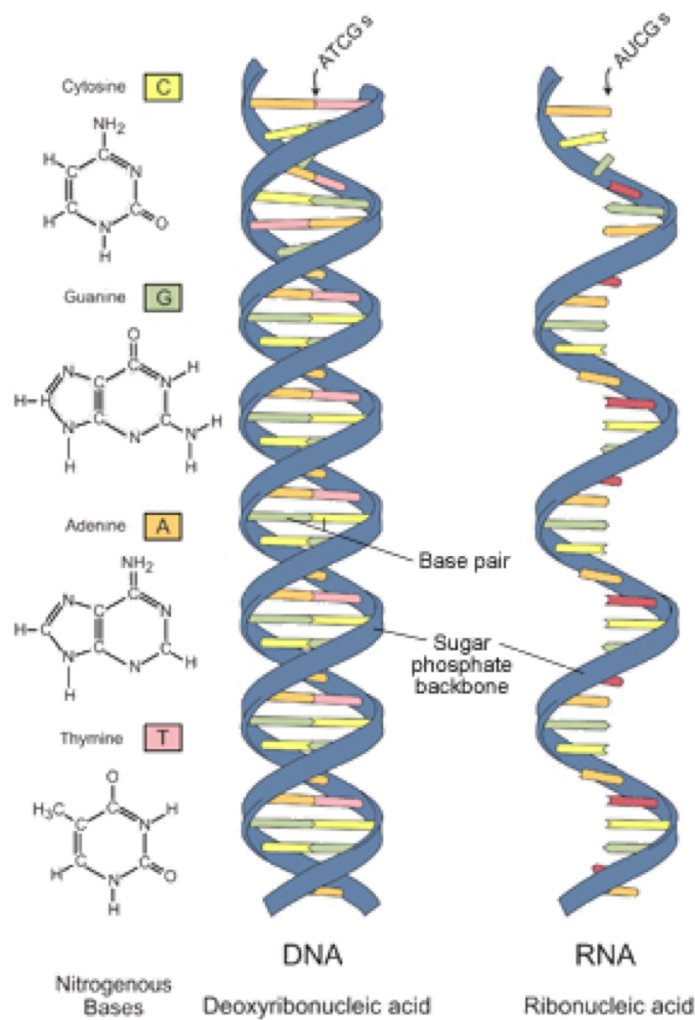
Flow off Genetic Information

- Genes direct protein synthesis through ribonucleic acid (RNA) intermediate
 - Proteins responsible for phenotypic traits
- **Central Dogma of Molecular Biology:**
 - Flow of genetic information is:
DNA $\xrightarrow{\text{transcription}}$ RNA $\xrightarrow{\text{translation}}$ Protein
 - DNA acts as template for RNA synthesis during process of transcription
 - RNA directs assembly of proteins (protein synthesis) during translation
 - Language of genetic code TRANSLATED to language of protein
 - AKA gene expression



RNA Structure and Function

- RNA: nucleic acid composed of nucleotides
- 4 basic differences between structure of DNA and RNA:
 1. RNA contains sugar ribose; DNA contains sugar deoxyribose
 2. RNA contains base uracil; DNA contains base thymine (U pairs with A just as T does)
 3. RNA is single stranded; DNA is double stranded
 1. RNA can base pair to itself in segments and form short pieces that are double-stranded
 4. RNA is short, usually length of one gene; DNA is very long, contains thousands of genes
- Cells contain 3 main types of RNA, each with a different function:
 - Messenger RNA (mRNA) carries instructions from gene in nucleus to synthesize protein in cytosol
 - Ribosomal RNA (rRNA) structural component of ribosomes
 - Organelle in cell where protein synthesis occurs
 - Transfer RNA (tRNA) transfers amino acids to ribosome to make protein
 - Contains 3 specialized nucleotides called 'anticodon' that bind to mRNA during translation



Transcription

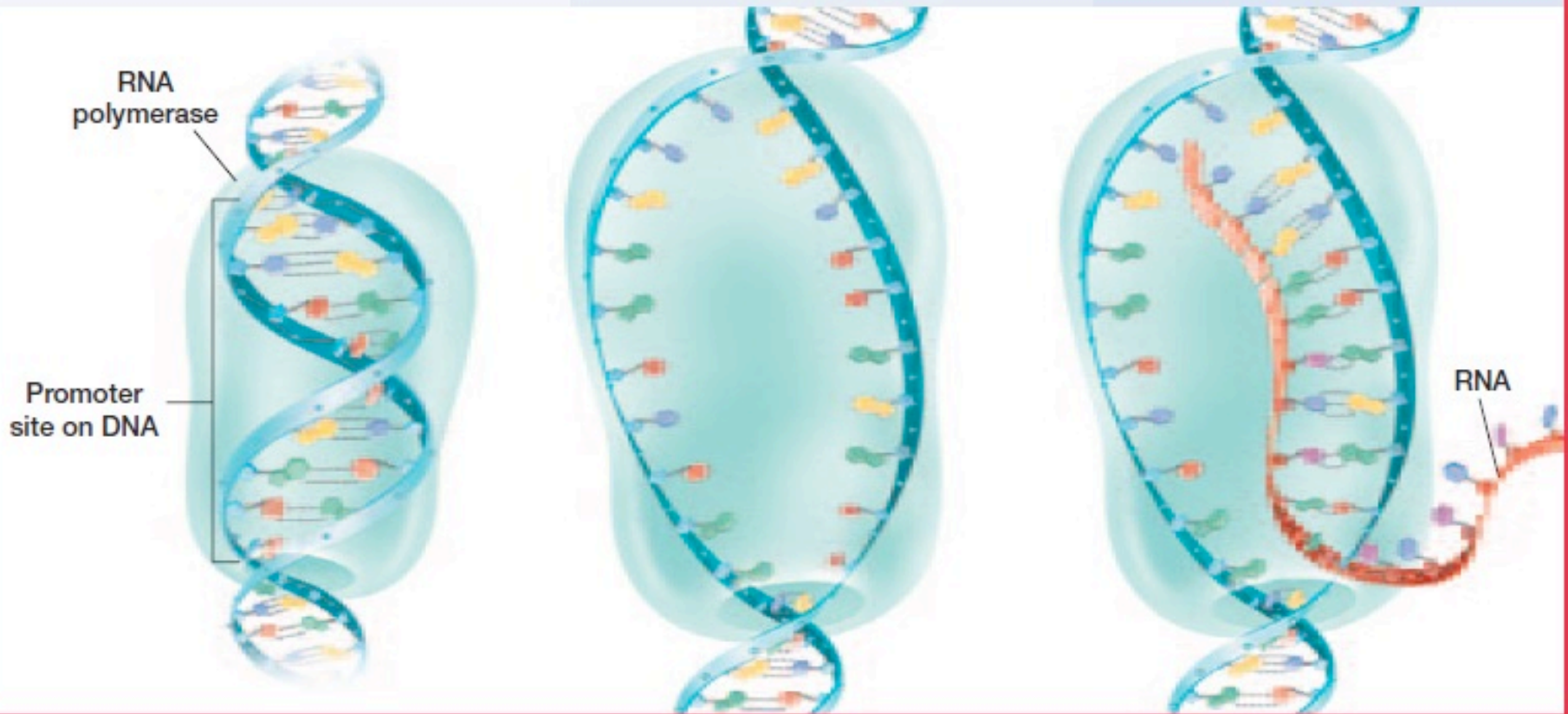
- Process by which genetic instructions are transcribed (re-written) into RNA molecule – produces mRNA Molecule
- Occurs in nucleus of eukaryotes; nucleoid region of prokaryotes
- 1. **RNA polymerase binds to promoter on DNA template molecule**
 - ▣ **Promoter:** specific DNA sequence that serves as recognition site for RNA polymerase to bind and initiate transcription
 - ▣ DNA strands are separated and unwound
- 2. **RNA polymerase adds free RNA nucleotides complementary to nucleotides on DNA template**
 - ▣ Complementary base pairing determines sequence of new RNA molecule based on DNA template
 - ▣ Only a specific region, gene, is transcribed. 5'→3' direction - elongation
 - ▣ One DNA strand used as template; as RNA polymerase moves past, DNA rewind
- 3. **RNA polymerase reaches termination signal; DNA sequence that marks end of gene**
 - ▣ Stop signal; RNA polymerase releases template DNA and newly synthesized RNA molecule
- New transcript can be started as soon as the promoter is free



Transcription: Making RNA

RNA polymerase adds complementary RNA nucleotides as it reads the gene.

- 1 RNA polymerase binds to the gene's promoter.
- 2 The two DNA strands unwind and separate.
- 3 Complementary RNA nucleotides are added.

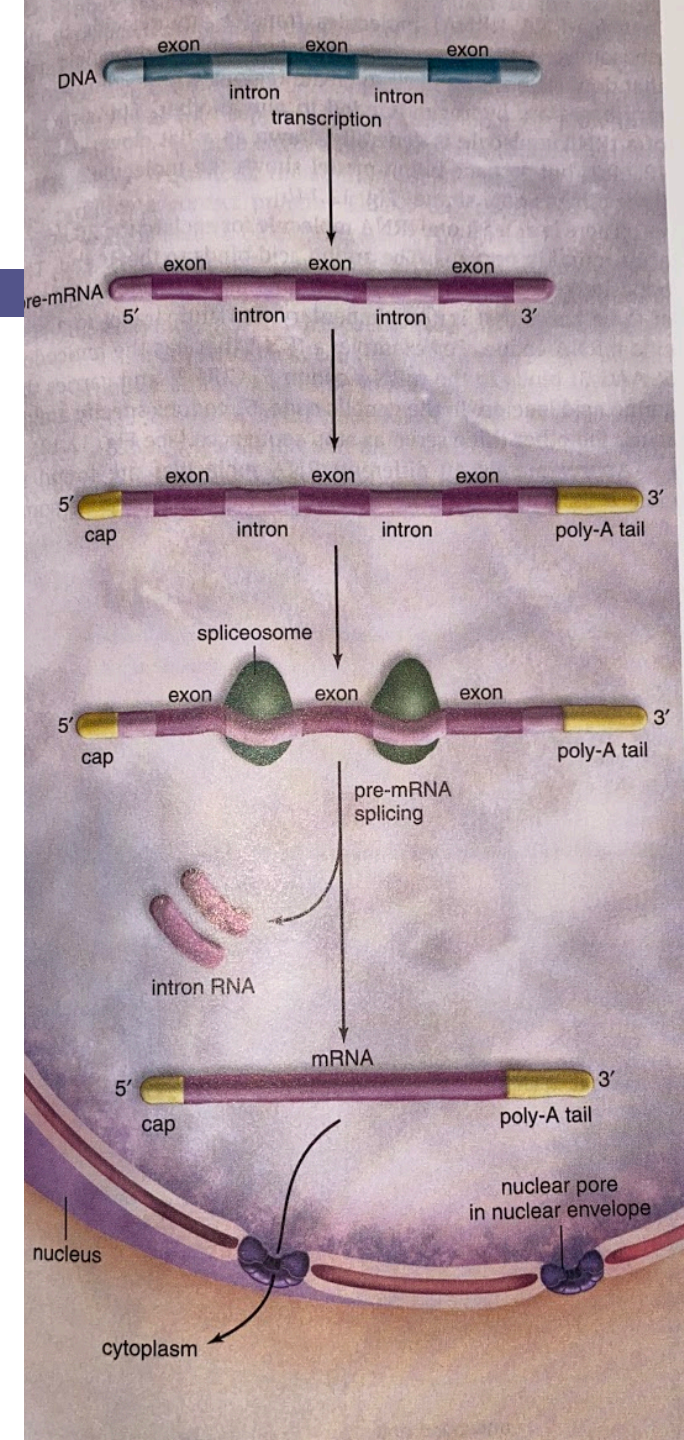


mRNA Processing

- Newly formed pre-mRNA must undergo processing before it leaves the nucleus
 - ▣ mRNA receives a **cap** at its 5' end
 - Modified Guanine. Tells a ribosome where to attached when translation begins
 - ▣ **Poly-A Tail** – 150-200 adenine (A) added to 3' end.
 - Facilitates transport out of nucleus
 - Helps initiate loading into ribosome
 - Delays degradation of mRNA by hydrolytic enzymes
 - ▣ Contains mix of **introns** and **exons**
 - **Intron** – Non-protein coding region of pre-mRNA
 - **Exon** – Protein coding region of pre-mRNA
 - *To create the final mature mRNA, a **spliceosome** will cut out the intron regions*

Introns

- Role was unknown for a long time
 - ▣ **Alternative mRNA Splicing**
 - One mRNA can lead to multiple different products depending on which exons are left in or out of final mRNA product.
 - ▣ **microRNAs**
 - Small molecules involved in regulating translation of mRNAs
 - Will bind with mRNA and prevent translation
- ▣ May play roll in **crossing-over** in meiosis – **exon shuffling**



Genetic Code

- Gene expression: amino acids assembled based on instructions encoded in sequence of nucleotides in mRNA
- **Genetic code:** rules that relate how sequence of bases in nucleotides correspond to particular amino acid
 - ▣ 3 adjacent nucleotides in mRNA specify an amino acid in a polypeptide
 - Known as a codon
 - ▣ 1 codon known as START; always found at the beginning of mRNA sequence; codes for amino acid methionine
 - **AUG**
 - ▣ 3 codons known as STOP; signifies end of polypeptide molecule
 - UAA, UAG, UGA
- Code is nearly universal among organisms
 - ▣ Supports the idea all organisms came from common ancestor
- 61 codons code for 20 amino acids + 3 STOP
- No codon codes for more than one amino acid; some amino acids encoded by 2 or more codons
 - ▣ Code is unambiguous and redundant

Genetic Codon Chart

	U	C	A	G	
U	UUU Phe	UCU Ser	UAU Tyr	UGU Cys	U
	UUC Phe	UCC Ser	UAC Tyr	UGC Cys	C
	UUA Leu	UCA Ser	UAA Stop	UGA Stop	A
	UUG Leu	UCG Ser	UAG Stop	UGG Trp	G
C	CUU Leu	CCU Pro	CAU His	CGU Arg	U
	CUC Leu	CCC Pro	CAC His	CGC Arg	C
	CUA Leu	CCA Pro	CAA Gln	CGA Arg	A
	CUG Leu	CCG Pro	CAG Gln	CGG Arg	G
A	AUU Ile	ACU Thr	AAU Asn	AGU Ser	U
	AUC Ile	ACC Thr	AAC Asn	AGC Ser	C
	AUA Ile	ACA Thr	AAA Lys	AGA Arg	A
	AUG Met	ACG Thr	AAG Lys	AGG Arg	G
G	GUU Val	GCU Ala	GAU Asp	GGU Gly	U
	GUC Val	GCC Ala	GAC Asp	GGC Gly	C
	GUA Val	GCA Ala	GAA Glu	GGA Gly	A
	GUG Val	GCG Ala	GAG Glu	GGG Gly	G

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Translation START codon

Translation STOP codon

Positively charged amino acids

Negatively charged amino acids

Hydrophobic amino acids

Hydrophilic non-charged amino acids

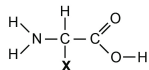
Cysteine

Proteins

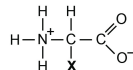
- Proteins made of one or more polypeptides
- Chains of amino acids linked by peptide bonds
- 20 different amino acids present in organisms
- Each polypeptide can be hundreds to thousands of amino acids in length
- Sequence of amino acids is unique to each protein
- Sequence of amino acids determines how protein will fold into its 3-dimensional structure
 - ▣ Structure dictates function

The Amino Acids

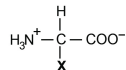
General Structure



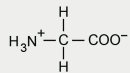
At Cellular pH



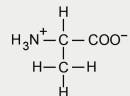
Shorthand



Glycine (GLY) [G]

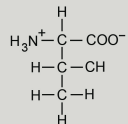


Alanine (Ala) [A]

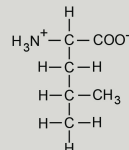


Non-polar, small

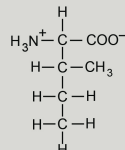
Valine (Val) [V]



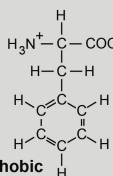
Leucine (Leu) [L]



Isoleucine (Ile) [I]

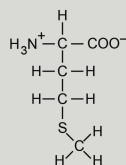


Phenylalanine (Phe) [F]

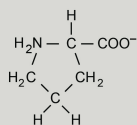


Non-polar, hydrophobic

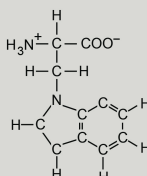
Methionine (Met) [M]



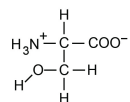
Proline (Pro) [P]



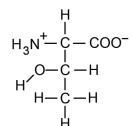
Tryptophan (Trp) [W]



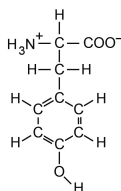
Serine (Ser) [S]



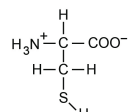
Threonine (Thr) [T]



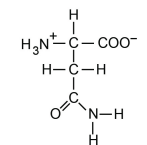
Tyrosine (Tyr) [Y]



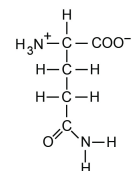
Cysteine (Cys) [C]



Asparagine (Asp) [N]

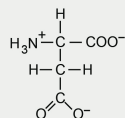


Glutamine (Gln) [Q]



Polar, neutral

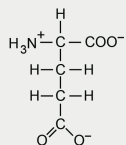
Aspartic Acid (Asp) [D]



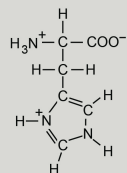
Acidic

Polar, charged

Glutamic Acid (Glu) [E]

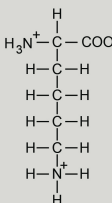


Histidine (His) [H]

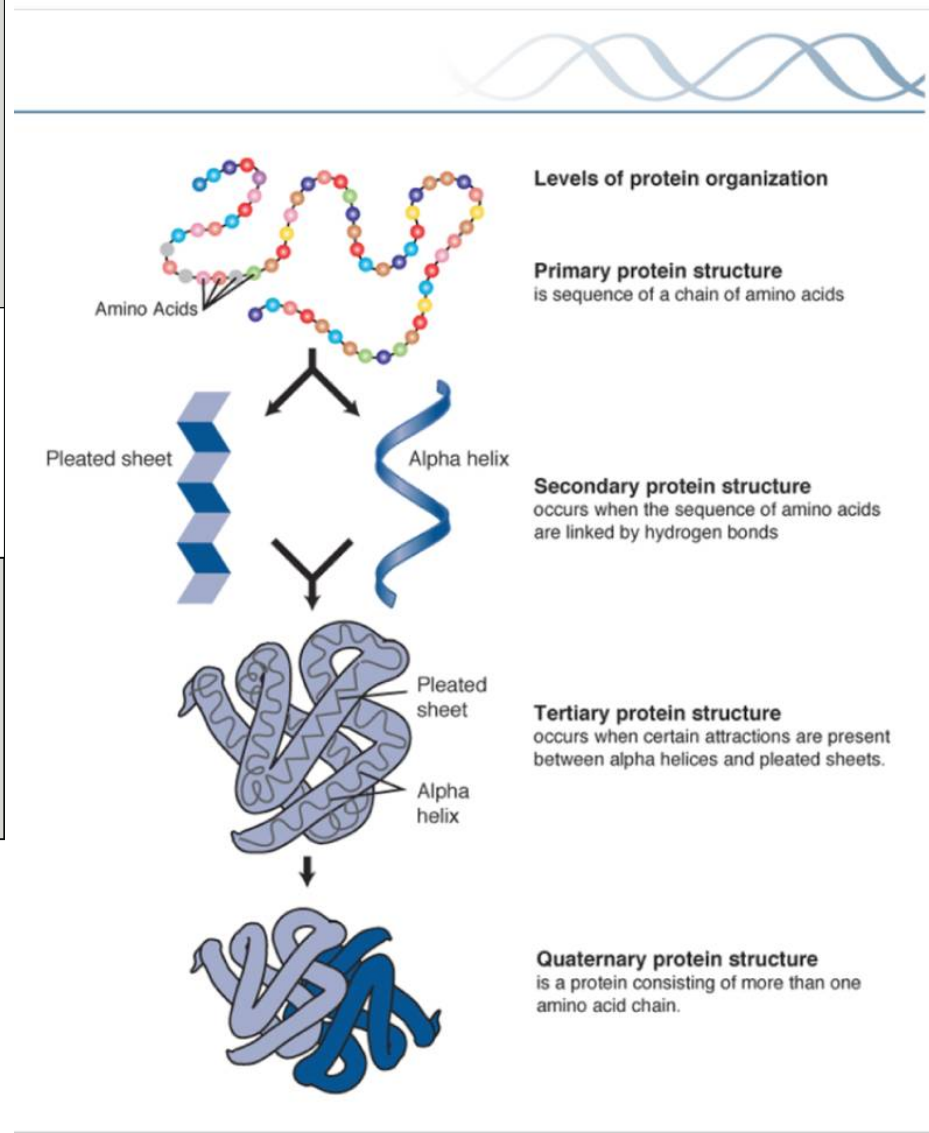
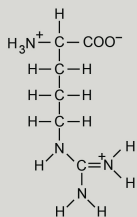


Basic

Lysine (Lys) [K]

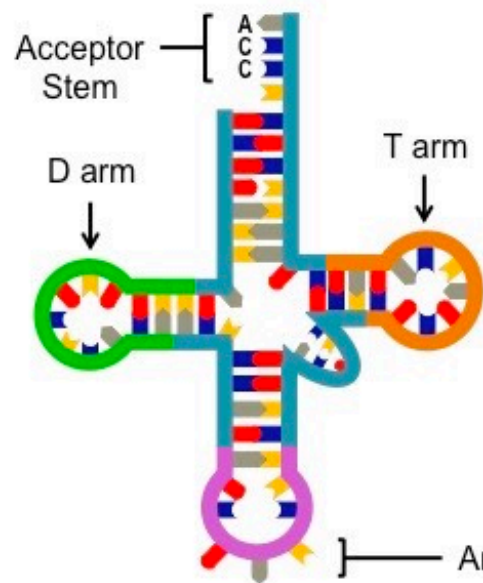


Arginine (Arg) [R]

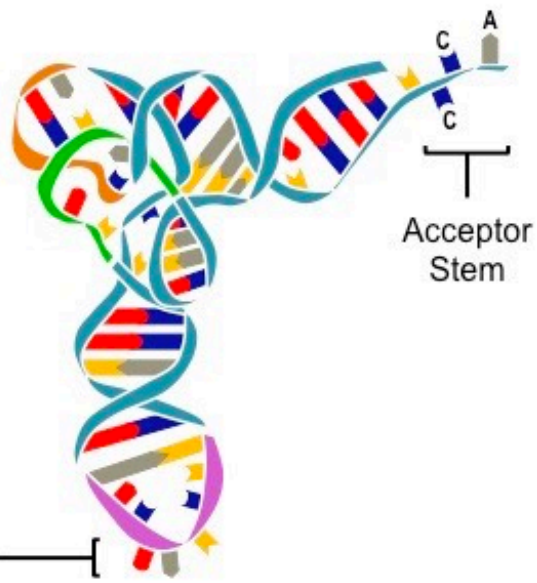


Translation – Role of tRNA

- tRNA molecules transfer amino acids to the ribozomes.
 - ▣ tRNA - flat, single stranded folded structure resembling a clover
- There is at least one tRNA for each of the 20 amino acids
 - ▣ Amino acid binds at the 3' end off the tRNA molecule
 - ▣ tRNA anticodon – region that is complementary to the 5'→3' codon on the mRNA strand
- Fewer tRNA molecules than codons (only 40)
 - ▣ Francis Crick – **Wobble Hypothesis**
 - ▣ First two positions of tRNA molecule can recognize up to 4 codons. Differ only by 3rd position.
 - ▣ Wobble allows for correct sequence to create protein
 - One reason genetic code is said to be **degenerate**
- Enzymes help with linking the tRNA to amino acids



2° Structure



3° Structure



Spacefill Model

Genetic Codon Chart

	U	C	A	G	
U	UUU Phe	UCU Ser	UAU Tyr	UGU Cys	U
	UUC Phe	UCC Ser	UAC Tyr	UGC Cys	C
	UUA Leu	UCA Ser	UAA Stop	UGA Stop	A
	UUG Leu	UCG Ser	UAG Stop	UGG Trp	G
C	CUU Leu	CCU Pro	CAU His	CGU Arg	U
	CUC Leu	CCC Pro	CAC His	CGC Arg	C
	CUA Leu	CCA Pro	CAA Gln	CGA Arg	A
	CUG Leu	CCG Pro	CAG Gln	CGG Arg	G
A	AUU Ile	ACU Thr	AAU Asn	AGU Ser	U
	AUC Ile	ACC Thr	AAC Asn	AGC Ser	C
	AUA Ile	ACA Thr	AAA Lys	AGA Arg	A
	AUG Met	ACG Thr	AAG Lys	AGG Arg	G
G	GUU Val	GCU Ala	GAU Asp	GGU Gly	U
	GUC Val	GCC Ala	GAC Asp	GGC Gly	C
	GUA Val	GCA Ala	GAA Glu	GGA Gly	A
	GUG Val	GCG Ala	GAG Glu	GGG Gly	G

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Translation START codon

Translation STOP codon

Positively charged amino acids

Negatively charged amino acids

Hydrophobic amino acids

Hydrophilic non-charged amino acids

Cysteine

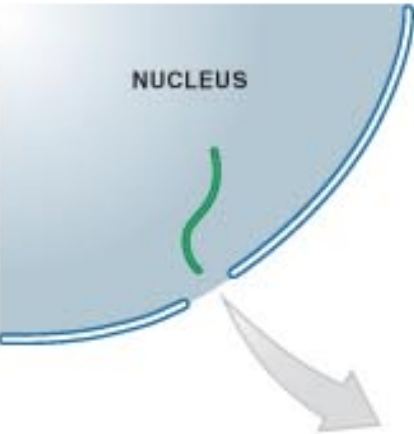
Translation – Role of rRNA/Ribosomes

- rRNA codes for 2 ribosomal subunits
 - ▣ Large and small subunit make up all ribosomes.
 - ▣ Join together at start of translation
 - ▣ Ribosomes can remain in cytoplasm or attached to ER
- Ribosome has a binding site for mRNA, and 3 binding sites for tRNA
 - ▣ E, P, & A sites in large subunit
 - E – Exit – location where the used tRNA exits
 - P – Protein – where tRNA bound to amino acid is processed to elongate the protein chain
 - A – Amino Acid – where next amino acid anticodon enters and waits to move to P site
- Enzyme Ribozyme helps create peptide bond between neighboring amino acids

Translation

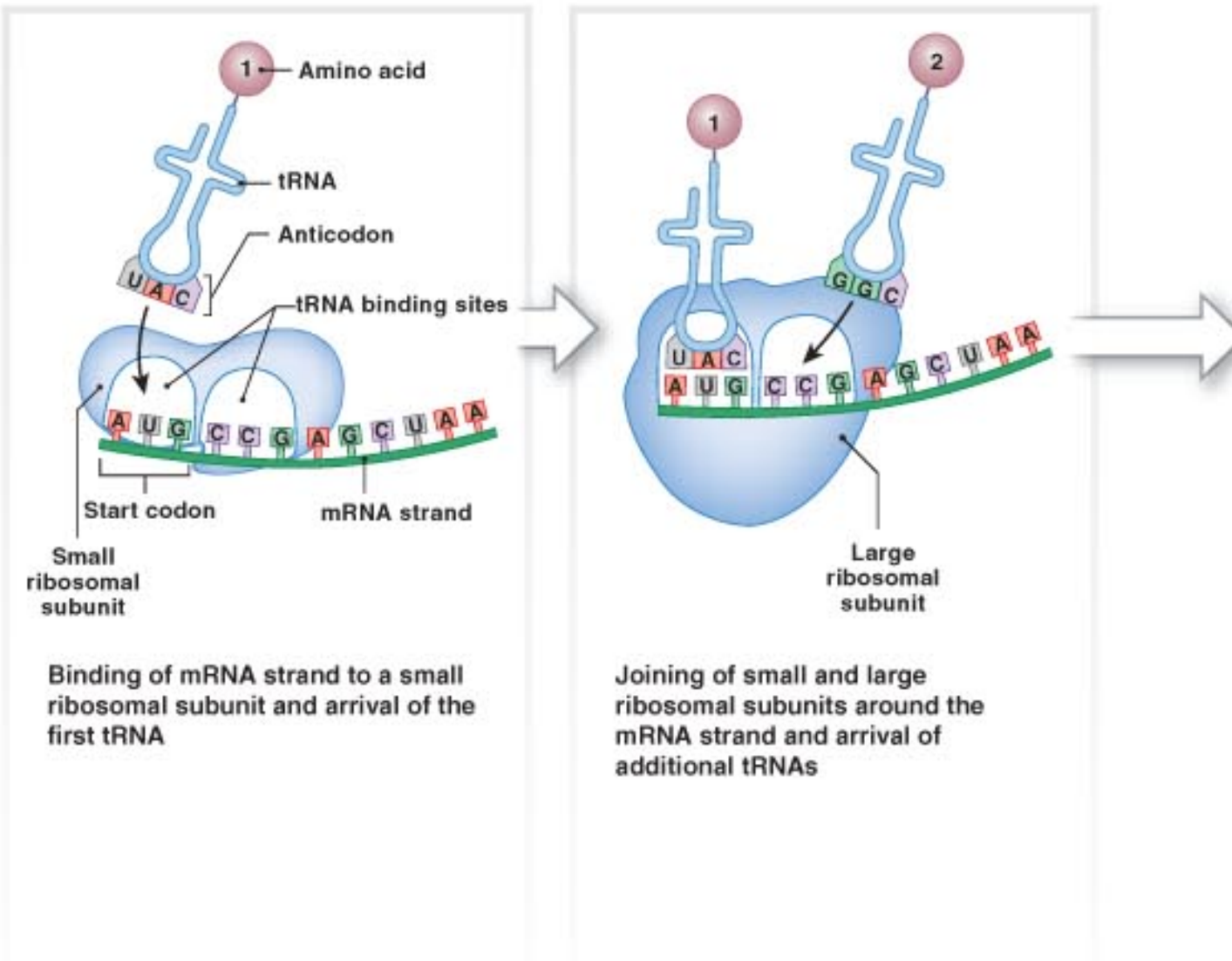
- Process by which protein is constructed from mRNA instructions; uses all forms of RNA; occurs in cytosol
- 1. **Initiation** - 2 ribosomal subunits, tRNA, mRNA join together
 - ▣ Enzymes attach corresponding amino acid to end of each tRNA molecule according to genetic code
 - ▣ Opposite end of tRNA contains anticodon: 3 nucleotide sequence that is complementary to codon in mRNA
 - ▣ Ex: tRNA with anticodon UAC will bind to mRNA codon AUG and carry amino acid methionine (start)
- 2. Next tRNA pairs its anticodon with mRNA codon
 - ▣ Ribosome detaches methionine from first tRNA; forms peptide bond between it and next amino acid
 - ▣ First tRNA leaves ribosome
 - ▣ Ribosome moves to next codon in sequence
- 3. **Elongation** - Polypeptide chain continues to grow
 - ▣ tRNAs come in carrying amino acids
 - ▣ **Translocation** - Growing chain transferred from one tRNA to amino acid attached to next tRNA
- 4. **Termination** - Ribosome reaches STOP codon; newly synthesized polypeptide is released
- 5. Translational components come apart; free to translate another mRNA

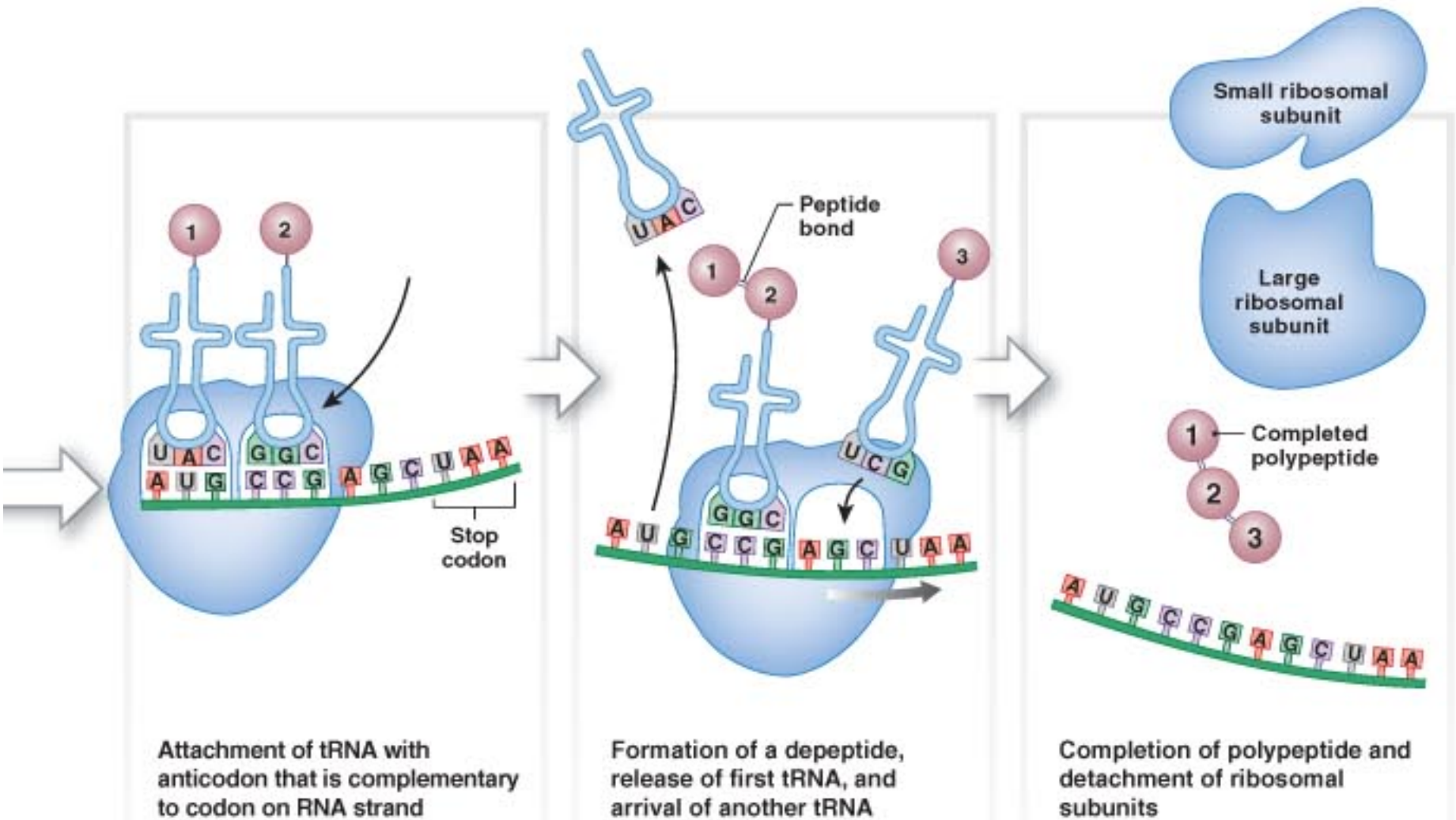
The process of translation



Entry of mRNA into cytoplasm

- A** Adenine
- G** Guanine
- C** Cytosine
- U** Uracil





The Human Genome

- Entire sequence of **human genome** is known
 - ▣ All 3.2 billion bases and 23 chromosomes
- How do we compute that much data and pull information out of it?
- **Bioinformatics**: field that applies computer science to biological systems
 - ▣ Can quickly compare many different DNA sequences
 - ▣ Can interpret DNA sequence and find the location of a gene on a chromosome
- Still working on determining where/when cells use proteins coded in the ~20-30,000 genes in the genome
- Why?
- Knowing the gene sequence that controls a biological function can help diagnose, treat, prevent genetic disorders, cancer, infectious diseases

1

5

Hypertension, essential, susceptibility to
 CLL/lymphoma, B-cell
 Lymphoma, diffuse large cell
 Necdin
 Prader-Willi syndrome
 Angelman syndrome
 Hair color, brown
 Spastic paraplegia
 Limb deformity
 Schizophrenia, neurophysiologic defect in
 Isovalericacidemia
 Spherocytosis, hereditary, Japanese type
 Bartter syndrome
 Amyotrophic lateral sclerosis, juvenile recessive
 Dyserythropoietic anemia, congenital, type III
 Griscelli syndrome
 Deafness, autosomal recessive
 Hepatic lipase deficiency
 Marfan syndrome
 Shprintzen-Goldberg syndrome
 Ectopia lentis, familial
 Leukemia, acute promyelocytic, PML/RARA type
 Cardiomyopathy, familial hypertrophic
 Enhanced S-cone syndrome
 Glutaricaciduria, type IIA
 Epilepsy, nocturnal frontal lobe, type 2
 PAPA syndrome
 Diabetes mellitus, insulin-dependent

100 million base pairs



Prader-Willi/Angelman syndrome (paternally imprinted)
 Eye color, brown
 Human coronavirus sensitivity
 Albinism, oculocutaneous, type II and ocular
 Andermann syndrome
 Cardiomyopathy, dilated and familial hypertrophic
 Epilepsy, juvenile myoclonic
 Spinocerebellar ataxia
 Microcephaly, primary autosomal recessive
 Dyserythropoietic anemia, congenital, type I
 Muscular dystrophy, limb-girdle, type 2A
 Dyslexia
 Amyloidosis, hemodialysis-related
 Ceroid-lipofuscinosis, neuronal, late infantile
 Gynecomastia, familial
 Virilization, maternal and fetal
 Colorectal cancer
 Carbohydrate-deficient glycoprotein syndrome, type Ib
 Bardet-Biedl syndrome
 Tay-Sachs disease
 GM2-gangliosidosis
 Tyrosinemia, type I
 Mental retardation, severe
 Hypercholesterolemia, familial, autosomal recessive
 Retinitis pigmentosa, autosomal recessive
 Otosclerosis
 Bloom syndrome